

MAT 310, Handout 8: Geometric Distribution, Negative Binomial Distribution (3.5, 3.6)

Geometric RV: Given independent trials where each trial has two outcomes, success and failure, with probabilities p and $q = 1 - p$, respectively. Let $Y \sim \text{Geom}(p)$ be the trial on which the 1st success occurs. The distribution of Y is given by:

$$p(y) =$$

Negative Binomial RV: Given independent trials where each trial has two outcomes, success and failure, with probabilities p and $q = 1 - p$, respectively. Let $Y \sim \text{NegBin}(r, p)$ be the trial on which the r^{th} success occurs. The distribution of Y is given by:

$$p(y) =$$

Remark: Geometric RV is a _____ RV with _____.

Example 1. Suppose a volleyball player serves an ace on 15% of their serves. Assuming each serve is independent, find the probability distribution for X , the number of the serve on which the player hits her first ace.

Example 2. Suppose you have a 40% probability of beating your roommate in ping pong. Assuming each game is independent, what is the probability that

(a) your first win occurs on the 3rd game.

(b) you win at least one of the first 4 games.

(c) your 3rd win occurs on the 10th game.

Expected Value of $Y \sim \text{Geom}(p)$

$$E(Y) =$$

Proof:

Variance of $Y \sim \text{Geom}(p)$

$$V(Y) =$$

Expected Value and Variance of $Y \sim \text{NegBin}(r, p)$

$$E(Y) = \qquad V(Y) =$$

Memoryless Property of Geometric RVs: If $Y \sim \text{Geom}(p)$ and $a, b > 0$, then

$$P(Y > a + b | Y > a) =$$

